

The normalization obstruction in prime-time averages

Conjecture 00000001695

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Abstract

Use the constant observable on the unit circle rotated by $\sqrt{2}$. This is a measure-preserving linear polynomial phase with a badly approximable coefficient. With the source's normalization by N , its prime-time average stays at least $1/4$ away from its integral for every $N \geq 16$. The asserted rate fails for every nonnegative constant at arbitrarily large N .

1 An admissible rotation

Let $X = \mathbb{R}/\mathbb{Z}$ with normalized Haar measure μ , let $T(x) = x + \sqrt{2} \pmod{1}$, and let $f(x) = 1$. Then T preserves μ and $T^p(x) = x + p\sqrt{2} \pmod{1}$, a linear polynomial phase. In particular $\int_X f d\mu = 1$. The coefficient satisfies the following explicit bad-approximation condition.

Lemma 1. *For every $p \in \mathbb{Z}$ and integer $q \geq 1$,*

$$\left| \sqrt{2} - \frac{p}{q} \right| \geq \frac{1}{5q^2}.$$

Proof. If the left side is at least one, the assertion follows from $q \geq 1$. Otherwise $|p/q| < \sqrt{2} + 1 < 3$, so $|\sqrt{2} + p/q| < 5$. The integer $2q^2 - p^2$ is nonzero: equality would make $\sqrt{2}$ rational. Consequently

$$1 \leq |2q^2 - p^2| = q^2 \left| \sqrt{2} - \frac{p}{q} \right| \left| \sqrt{2} + \frac{p}{q} \right| < 5q^2 \left| \sqrt{2} - \frac{p}{q} \right|.$$

This gives the required inequality. □

2 The prime-counting obstruction

Write $\pi(N) = \#\{p \leq N : p \text{ is prime}\}$. The average in the source is

$$A_N(x) = \frac{1}{N} \sum_{p \leq N} f(T^p x) = \frac{\pi(N)}{N}.$$

An elementary sieve using only the parity of primes gives the convenient bound $2\pi(N) \leq N + 4$ for $N \geq 2$. Thus for $N \geq 16$,

$$\frac{\pi(N)}{N} \leq \frac{1}{2} + \frac{2}{N} \leq \frac{3}{4}, \quad \left| A_N(x) - \int_X f d\mu \right| \geq \frac{1}{4}.$$

No prime number theorem or quantitative distribution theorem is needed.

3 Every proposed constant fails

Theorem 1. *For every $C \geq 0$, every natural threshold N_0 , and every $x \in X$, there is an integer $N \geq \max(N_0, 16)$ such that*

$$C \frac{\log \log N}{\sqrt{N} \log N} < \left| A_N(x) - \int_X f d\mu \right|.$$

Proof. For $N \geq 2$, $\log N > 0$ and $\log(\log N) \leq \log N$. Therefore

$$C \frac{\log \log N}{\sqrt{N} \log N} \leq \frac{C}{\sqrt{N}}.$$

Choose an integer $N > (4C+1)^2 + N_0 + 16$. Then $N \geq N_0$, $N \geq 16$, and $\sqrt{N} > 4C$, so $C/\sqrt{N} < 1/4$. Apply the preceding uniform lower bound on the error. $\square$

Thus no nonnegative constant works even eventually. The observable is smooth and bounded, so regularity cannot repair this estimate.

Scope. The counterexample uses the stated normalization $1/N$ and integral. There is no zero-mean hypothesis. A $\pi(N)$ normalization or logarithmic prime weights give different expressions, outside this disproof's scope.

4 Formal verification

Lean uses the actual `UnitAddCircle`, Haar volume, Bochner integral, function iteration, prime finsets, square root, and logarithm.

- `sqrt_two_badly_approximable` proves the explicit $1/(5q^2)$ bound. The rotation preserves measure and its iterates are the displayed linear phase.
- `actual_average` and `actual_integral` compute the prime sum and integral. `prime_count_bound` uses the sieve with modulus two; `error_lower_bound` proves the resulting uniform error bound.
- `arbitrarily_large_failure` proves the theorem for every $C \geq 0$, threshold, and point. The final `counterexample` combines it with the actual rotation and bad-approximation property.

Only standard logical axioms are used, with no unproved placeholders, custom axioms, native evaluation, or numerical experiments.

Reproduction. In `lean/`, run:

```
lake build
lake env lean -DwarningAsError=true Main.lean
```

Lean 4.19.0 and an exact Mathlib revision are pinned. Run `tectonic main.tex` to rebuild this PDF.

Sources. Conjecture 00000001695; Mathlib c44e0c8ee63c.